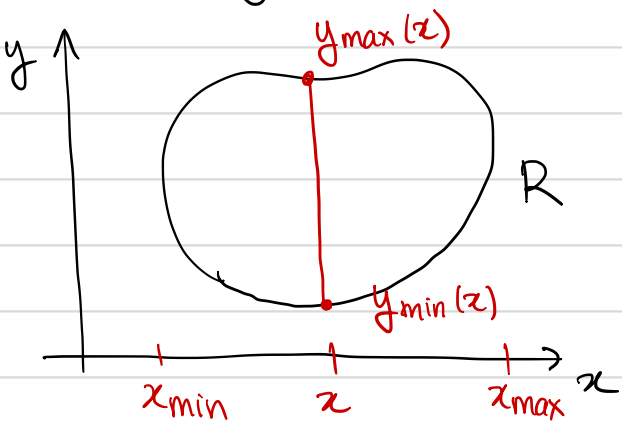
Lecture 25

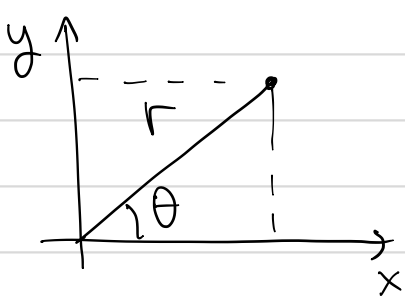
- Course review

Double integrals



$$\int_{x_{min}}^{x_{max}} \int_{y_{min}(x)}^{y_{max}(x)} f(x,y)\, dy\, dx$$

Polar coordinates

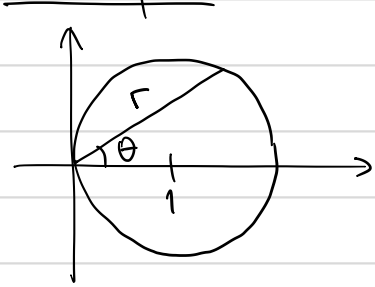


$x = r\cos\theta$

$y = r\sin\theta$

$dA = r\,dr\,d\theta$

Example:



$(x-1)^2 + y^2 = 1$

$x^2 - 2x + 1 + y^2 = 1$

$r^2 - 2r\cos\theta = 0$

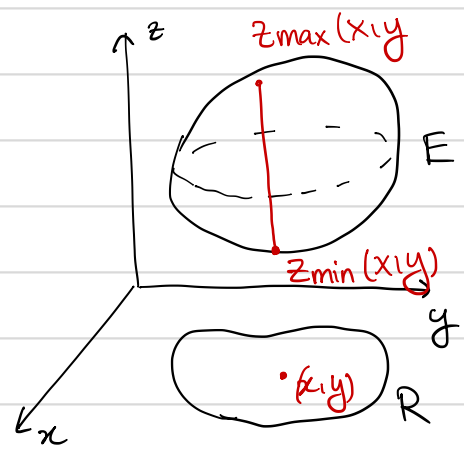
$$\begin{cases} r \leq 2\cos\theta \\ -\pi/2 \leq \theta \leq \pi/2 \end{cases}$$

General u-v coordinates

$$dx\,dy = \left| \frac{\partial(x,y)}{\partial(u,v)} \right| du\,dv$$

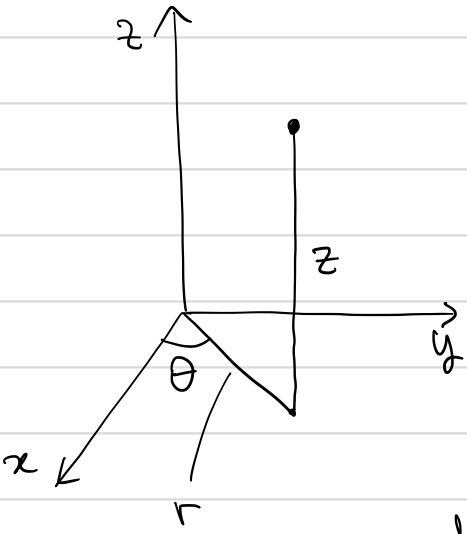
$$\frac{\partial(x,y)}{\partial(u,v)} = \det \begin{bmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{bmatrix}$$

Triple Integrals



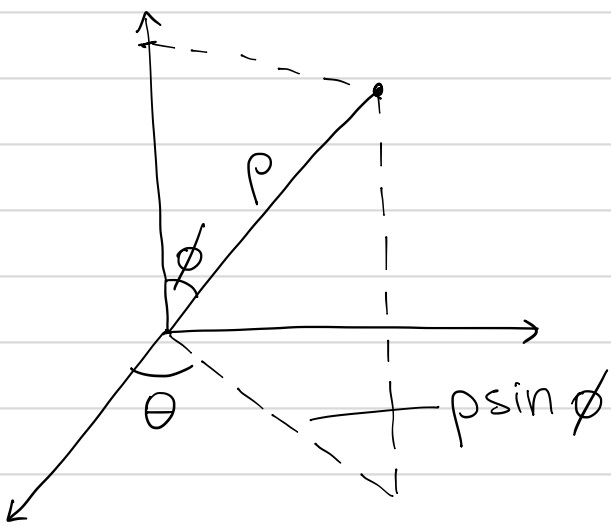
$$\iint_R \int_{z_{min}(x,y)}^{z_{max}(x,y)} f(x,y,z)\, dz\, dA$$

Cylindrical & spherical coords.



$$x = r\cos\theta$$
$$y = r\sin\theta$$
$$z = z$$

$$dV = r\,dr\,d\theta\,dz$$



$$z = \rho\cos\phi$$
$$x = \rho\sin\phi\cos\theta$$
$$y = \rho\sin\phi\sin\theta$$

$$dV = \rho^2\sin\phi\,d\rho\,d\phi\,d\theta$$

Chain rule

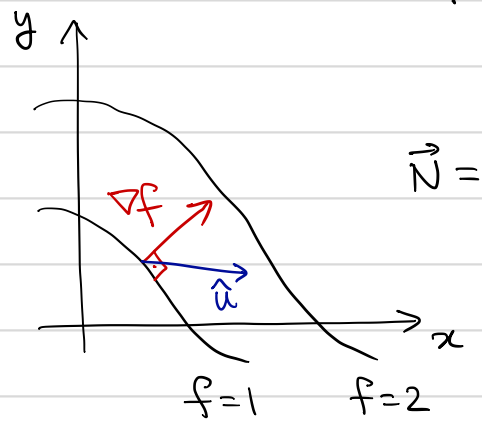
$$\Delta f \approx \frac{\partial f}{\partial x}\Delta x + \frac{\partial f}{\partial y}\Delta y$$

$$x = x(t)$$
$$y = y(t)$$

$$\frac{df}{dt} = \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt}$$

Gradient vector

$$\nabla f = \langle f_x, f_y \rangle$$



$$\vec{N} = \pm\frac{\nabla f}{|\nabla f|}$$

Directional derivative

$\vec{u}$ = unit vector

$$D_{\vec{u}} f = \nabla f \cdot \vec{u}$$

Work and line integrals

$\vec{F} = \langle P, Q, R \rangle$

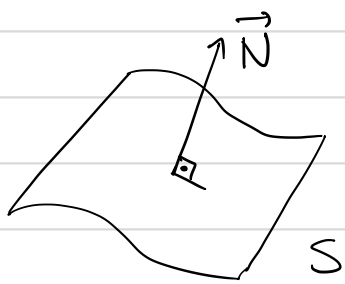
$$\int_C \vec{F} \cdot d\vec{r} = \int_C P\,dx + Q\,dy + R\,dz$$
$$= \int_C \vec{F} \cdot \vec{T}\,ds$$

$\vec{r}(t) = \langle x, y, z \rangle$

or x, or y, or time, or angle, ...

$\vec{T} = \dfrac{\vec{r}\,'(t)}{|\vec{r}\,'(t)|}$ $\qquad ds = |\vec{r}\,'(t)|\,dt$

Flux and surface integrals



$\vec{F} = \langle P, Q, R \rangle$

$$\iint_S \vec{F} \cdot d\vec{S} = \iint_S \vec{F} \cdot \hat{N}\,dS$$
$$= \iint_D \vec{F} \cdot (\vec{r}_u \times \vec{r}_v)\,du\,dv$$

$\vec{r}(u, v) = \langle x, y, z \rangle$

$\hat{N} = \pm \dfrac{\vec{r}_u \times \vec{r}_v}{|\vec{r}_u \times \vec{r}_v|}$ $\qquad dS = |\vec{r}_u \times \vec{r}_v|\,du\,dv$

Graphs: $S: \; z = f(x, y)$

$$\Rightarrow \vec{N} = \pm \langle -f_x, -f_y, 1 \rangle$$

$$\hat{N} = \frac{\langle -f_x, -f_y, 1 \rangle}{\sqrt{f_x^2 + f_y^2 + 1}}$$

$$dS = \sqrt{f_x^2 + f_y^2 + 1}\,dx\,dy$$

Div and curl

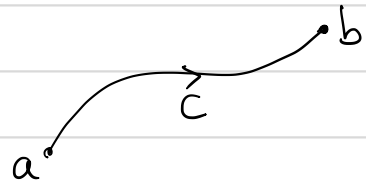
$\vec{F} = \langle P, Q, R \rangle$

$$\operatorname{div} \vec{F} = P_x + Q_y + R_z$$
$$= \nabla \cdot \vec{F}$$

$$\operatorname{curl} \vec{F} = \nabla \times \vec{F}$$
$$= \langle R_y - Q_z, P_z - R_x, Q_x - P_y \rangle$$

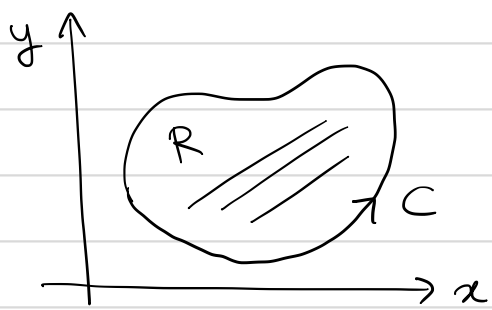
Fund. thm. of calculus for line integrals

$$f(b) - f(a) = \int_C \nabla f \cdot d\vec{r}$$



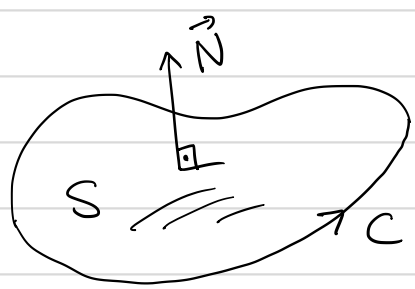
Green's theorem

$$\oint_C \vec{F}\cdot d\vec{r} = \iint_R (Q_x - P_y)\, dA$$



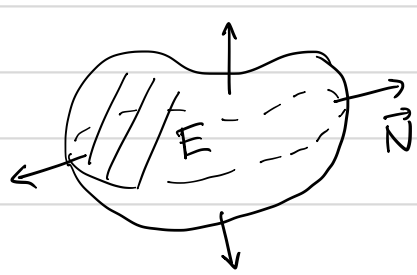
Stokes' theorem

$$\oint_C \vec{F}\cdot d\vec{r} = \iint_S (\nabla\times\vec{F})\cdot d\vec{S}$$



Divergence theorem

$$\oiint_S \vec{F}\cdot d\vec{S} = \iiint_E \operatorname{div}(\vec{F})\, dV$$



Conservative fields

If
- $\operatorname{curl}(\vec{F}) = 0$

[2D: 1 condition $Q_x = P_y$
3D: 3 conditions]

and
- $\vec{F}$ is defined on a <u>simply connected</u> region,

then
- $\vec{F} = \nabla f$ for some f.

How to find f:

$\vec{F} = \langle P, Q\rangle$

$$\frac{\partial f}{\partial x} = P \Rightarrow f = \int P\,dx + g(y)$$

$$\Rightarrow Q = \frac{\partial}{\partial y}\left[\int P\,dx\right] + g'(y)$$